LECTURE-25

SST Sum of Squares of Total

SSR (regression) explains variation by Target variable in results

SSE (error) explains variation by other factors than target variable.

tells kitnay percent ~~data~~ change in y can be explained by x (SSR) or other variables than x (SSE).

$$SST = SSR + SSE.$$

$$SST = \Sigma y^2 - \frac{(\Sigma y)^2}{n} = \Sigma (y - \bar{y})^2$$

$\hat{y} = 80 + 4x$

$$SSR = \Sigma (\hat{y} - \bar{y})^2$$

$$SSE = \Sigma y^2 - (c \cdot \Sigma y) - (m \Sigma xy)$$

$$R^2 = \text{Coff. of determination} = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$$

$$S_{y \cdot x} = \text{Standard error of Est.} = \sqrt{\frac{SSE}{n-2}}$$

⇒ ANOVA Table.

No. of Independent ways something can vary.

Source of variation	Sum of Squares	Degree of Freedom (DoF)	Mean Square	F
Regression	SS R	1 in simple linear regress	$MSR = \frac{SSR}{DOF}$	$F = \frac{MSR}{MSE}$
Error	SSE	n-2	$MSE = \frac{SSE}{DOF}$	
Total	SST	n-1		

→ Test Linearity of x & y in pop.

F-distribution table: Has two parameters i.e.:

1) DOF of Numerator = DOF of Regression
2) DOF of Denominator = DOF of Error.

- α values will be given (or 0.05)

⇒ Reject/Accept.

- Reject H_0 if: $\left(F = \frac{MSR}{MSE}\right) > \left(\text{Calculated value of F from the table}\right)$

$H_0 : \beta_1 = 0$ (No linear relationship b/w x & y when we run the eq. of regression line on entire population).

$H_1 : \beta_1 \neq 0$ (means there is linear relation b/w x & y).

probability

of α

⇒ P-value : Minimum value of α jispar Ho reject ho ~~vha~~ ho
* (It can only be found by software)

• Reject Ho if : $\boxed{\text{p-value} \le \alpha}$

T-Test for testing Linearity b/w x & y.

If value of Ho (i.e: 0) falls in range of 'b' then it will be accepted (non linear) else if not in the range then Ho rejected (linear relationship).

$$\boxed{b \pm \underbrace{t_{\alpha/2,(n-2)}}_{\text{from table}} * S_b}$$

$$S_b = \frac{S_{y.x}}{\sqrt{\sum y^2 - \frac{(\sum y)^2}{n}}} \qquad \frac{SSE}{n-2}$$